

```
In [1]: import numpy as np
```

```
In [2]: import matplotlib.pyplot as plt
```

```
In [4]: from PIL import Image#python image library
```

```
In [18]: img = Image.open(r'C:\Users\Tharuni\Desktop\image1.jpg')
```

```
In [19]: img
```

```
Out[19]:
```



```
In [20]: type(img)
```

```
Out[20]: PIL.JpegImagePlugin.JpegImageFile
```

```
In [21]: arr = np.array(img)  
arr
```

```

Out[21]: array([[ 94,  95,  89],
                [ 95,  94,  89],
                [ 95,  94,  89],
                ...,
                [ 25,  33,  36],
                [ 25,  33,  36],
                [ 24,  32,  35]],

                [[ 96,  97,  91],
                [ 96,  97,  91],
                [ 98,  97,  92],
                ...,
                [ 25,  33,  36],
                [ 25,  33,  36],
                [ 25,  33,  36]],

                [[ 98, 101,  94],
                [ 99, 100,  94],
                [100, 101,  95],
                ...,
                [ 25,  33,  36],
                [ 25,  33,  36],
                [ 25,  33,  36]],

                ...,

                [[ 78,  77,  75],
                [ 73,  73,  71],
                [ 69,  69,  67],
                ...,
                [ 45,  43,  44],
                [ 51,  49,  52],
                [ 83,  81,  84]],

                [[ 76,  75,  73],
                [ 71,  71,  69],
                [ 69,  69,  67],
                ...,
                [ 55,  53,  54],
                [ 75,  73,  76],
                [ 72,  70,  73]],

                [[ 73,  72,  70],
                [ 69,  69,  67],
                [ 67,  67,  65],
                ...,
                [ 59,  57,  58],
                [ 94,  92,  95],
                [ 81,  79,  82]]], dtype=uint8)

```

```

In [22]: img1 = Image.open(r'C:\Users\Tharuni\Desktop\car.jpg')
img1

```

Out[22]:

In [23]: `type(img1)`Out[23]: `PIL.JpegImagePlugin.JpegImageFile`In [24]: `img1_arr = np.asarray(img1)`
`img1_arr`

```

Out[24]: array([[ 85,  94,  93],
                [ 85,  94,  93],
                [ 85,  94,  93],
                ...,
                [234, 234, 232],
                [234, 234, 232],
                [234, 234, 232]],

                [[ 85,  94,  93],
                 [ 85,  94,  93],
                 [ 86,  95,  94],
                 ...,
                 [234, 234, 232],
                 [234, 234, 232],
                 [234, 234, 232]],

                [[ 86,  95,  94],
                 [ 86,  95,  94],
                 [ 87,  96,  95],
                 ...,
                 [234, 234, 232],
                 [234, 234, 232],
                 [234, 234, 232]],

                ...,

                [[ 86,  93,  99],
                 [ 79,  86,  92],
                 [ 84,  91,  97],
                 ...,
                 [ 57,  66,  71],
                 [ 65,  74,  79],
                 [ 69,  78,  83]],

                [[ 99, 106, 112],
                 [ 92,  99, 105],
                 [ 92,  99, 105],
                 ...,
                 [ 55,  64,  69],
                 [ 63,  72,  77],
                 [ 61,  70,  75]],

                [[ 62,  69,  75],
                 [ 74,  81,  87],
                 [ 94, 101, 107],
                 ...,
                 [ 51,  60,  67],
                 [ 58,  67,  74],
                 [ 49,  58,  65]]], dtype=uint8)

```

```
In [25]: type(img1_arr)
```

```
Out[25]: numpy.ndarray
```

```
In [26]: img1_arr.shape
```

```
Out[26]: (183, 276, 3)
```

```
In [27]: plt.imshow(img1_arr)
```

Out[27]: <matplotlib.image.AxesImage at 0x2d3c56364e0>



```
In [28]: img1_red = img1_arr.copy()
```

```
In [29]: img1_red == img1_arr
```

```
Out[29]: array([[ True,  True,  True],
               [ True,  True,  True],
               [ True,  True,  True],
               ...,
               [ True,  True,  True],
               [ True,  True,  True],
               [ True,  True,  True]],

              [[ True,  True,  True],
               [ True,  True,  True],
               [ True,  True,  True],
               ...,
               [ True,  True,  True],
               [ True,  True,  True],
               [ True,  True,  True]],

              [[ True,  True,  True],
               [ True,  True,  True],
               [ True,  True,  True],
               ...,
               [ True,  True,  True],
               [ True,  True,  True],
               [ True,  True,  True]],

              ...,

              [[ True,  True,  True],
               [ True,  True,  True],
               [ True,  True,  True],
               ...,
               [ True,  True,  True],
               [ True,  True,  True],
               [ True,  True,  True]],

              [[ True,  True,  True],
               [ True,  True,  True],
               [ True,  True,  True],
               ...,
               [ True,  True,  True],
               [ True,  True,  True],
               [ True,  True,  True]],

              [[ True,  True,  True],
               [ True,  True,  True],
               [ True,  True,  True],
               ...,
               [ True,  True,  True],
               [ True,  True,  True],
               [ True,  True,  True]])
```

```
In [30]: plt.imshow(img1_red)
```

```
Out[30]: <matplotlib.image.AxesImage at 0x2d3c569d9d0>
```

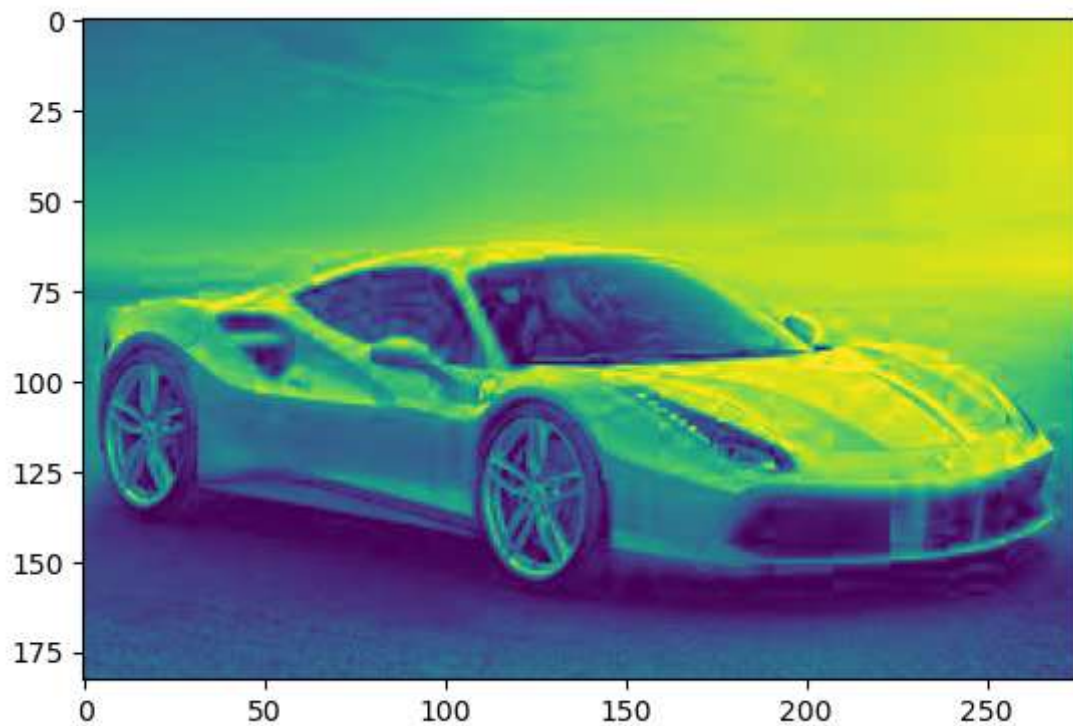


```
In [31]: img1_red.shape
```

```
Out[31]: (183, 276, 3)
```

```
In [32]: plt.imshow(img1_red[:, :, 0])# r g b
```

```
Out[32]: <matplotlib.image.AxesImage at 0x2d3c58e2a80>
```



```
In [33]: img1_red[:, :, 0]
```



```
Out[33]: array([[ 85,  85,  85, ..., 234, 234, 234],
                [ 85,  85,  86, ..., 234, 234, 234],
                [ 86,  86,  87, ..., 234, 234, 234],
                ...,
                [ 86,  79,  84, ...,  57,  65,  69],
                [ 99,  92,  92, ...,  55,  63,  61],
                [ 62,  74,  94, ...,  51,  58,  49]], dtype=uint8)
```

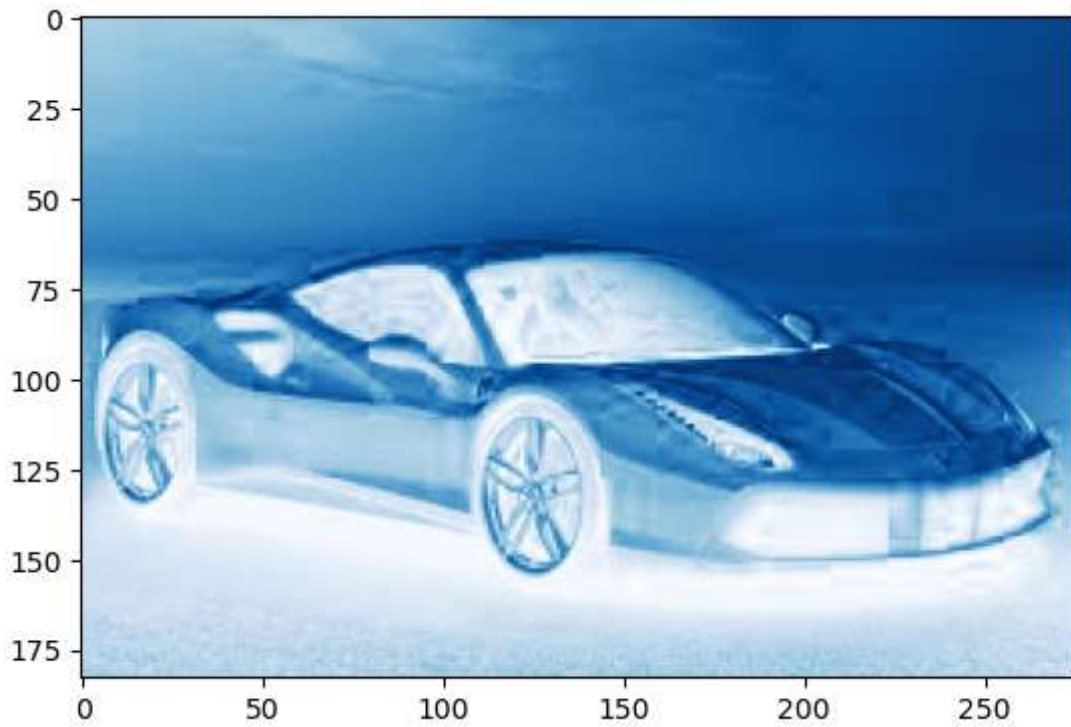
```
In [34]: plt.imshow(img1_red[:, :, 0], cmap = 'gray')
```

```
Out[34]: <matplotlib.image.AxesImage at 0x2d3c58ceba0>
```



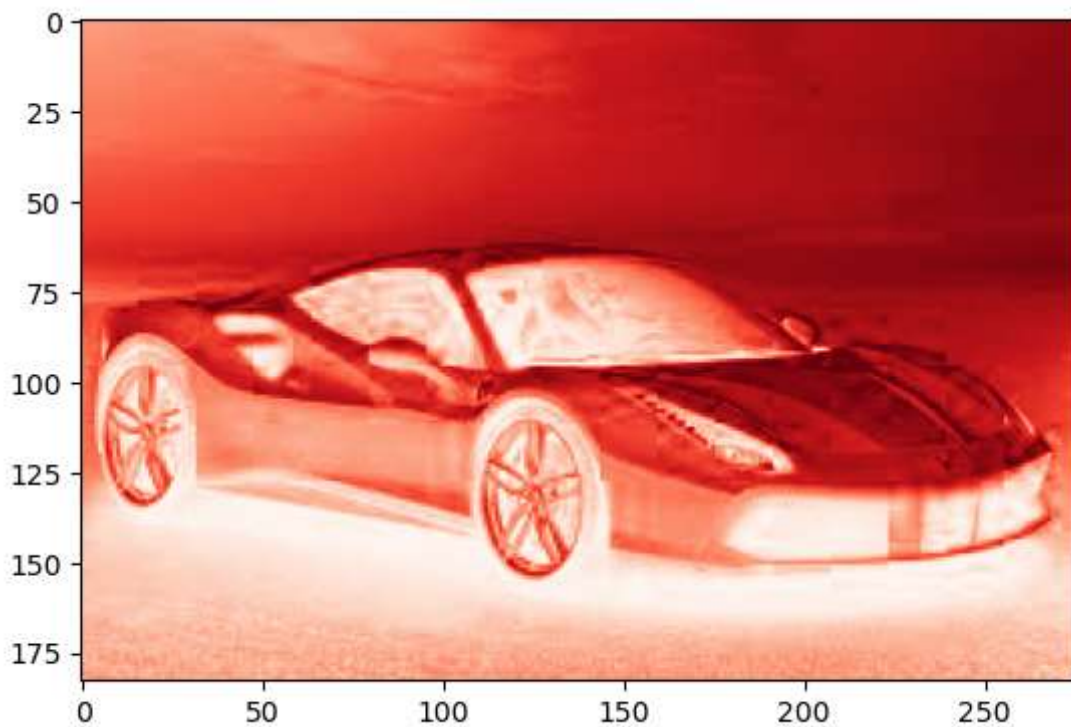
```
In [36]: plt.imshow(img1_red[:, :, 0], cmap = 'Blues')
```

```
Out[36]: <matplotlib.image.AxesImage at 0x2d3ca710c50>
```

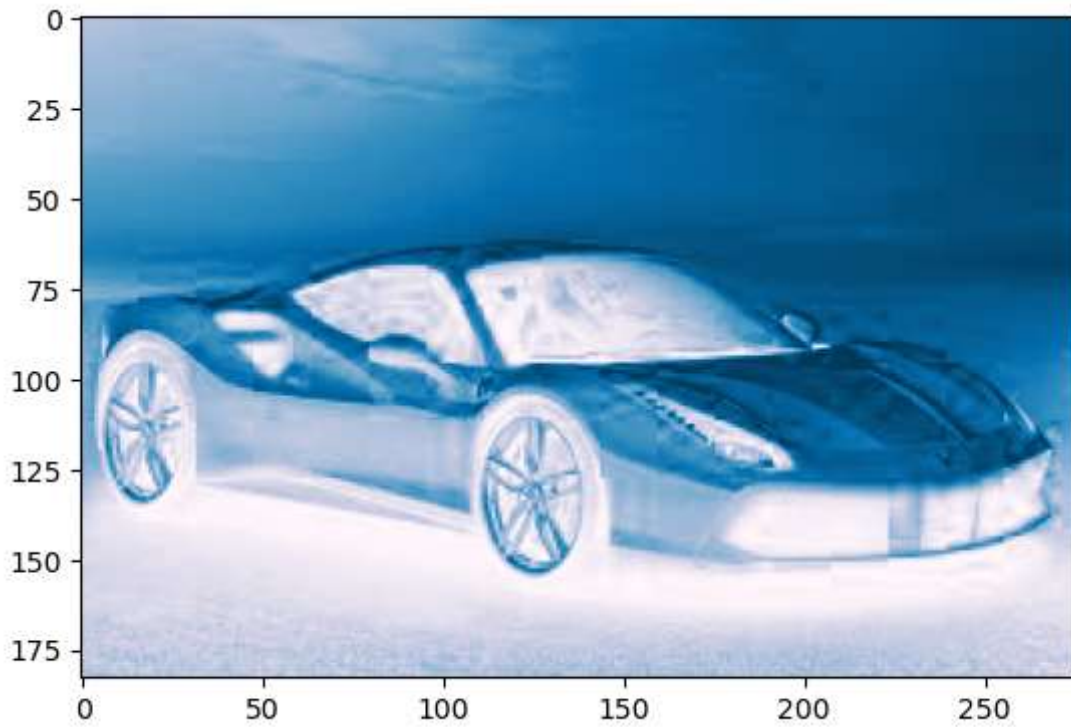
```
In [37]: plt.imshow(img1_red[:, :, 0], cmap = 'Reds')
```

```
Out[37]: <matplotlib.image.AxesImage at 0x2d3ca78d5e0>
```



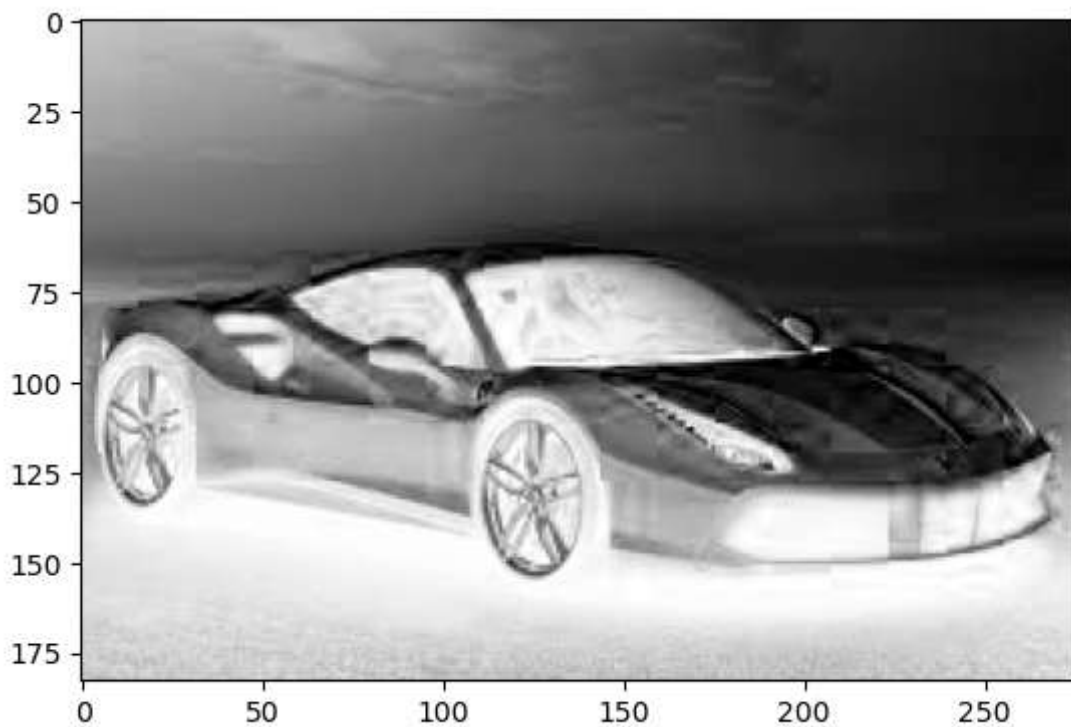
```
In [38]: plt.imshow(img1_red[:, :, 0], cmap = 'PuBu')
```

```
Out[38]: <matplotlib.image.AxesImage at 0x2d3c5832a20>
```



```
In [39]: plt.imshow(img1_red[:, :, 0], cmap = 'Greys')
```

```
Out[39]: <matplotlib.image.AxesImage at 0x2d3ca98baa0>
```



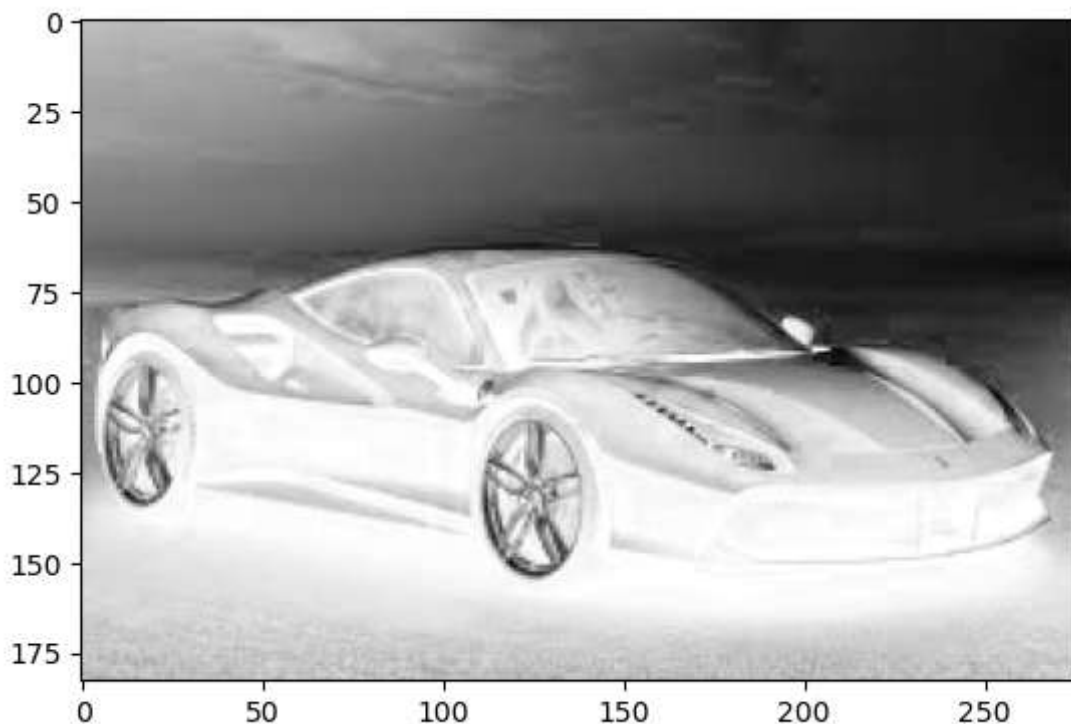
```
In [41]: plt.imshow(img1_red[:, :, 1], cmap = 'grey')
```

```
Out[41]: <matplotlib.image.AxesImage at 0x2d3c771e6f0>
```



```
In [43]: plt.imshow(img1_red[:, :, 2], cmap = 'Greys')
```

```
Out[43]: <matplotlib.image.AxesImage at 0x2d3ca84f6b0>
```



```
In [44]: img1_red[:, :, 0]
```

```
Out[44]: array([[ 85,  85,  85, ..., 234, 234, 234],
                [ 85,  85,  86, ..., 234, 234, 234],
                [ 86,  86,  87, ..., 234, 234, 234],
                ...,
                [ 86,  79,  84, ...,  57,  65,  69],
                [ 99,  92,  92, ...,  55,  63,  61],
                [ 62,  74,  94, ...,  51,  58,  49]], dtype=uint8)
```

```
In [45]: img1_red[:, :, 1]
```

```
Out[45]: array([[ 94,  94,  94, ..., 234, 234, 234],
                [ 94,  94,  95, ..., 234, 234, 234],
                [ 95,  95,  96, ..., 234, 234, 234],
                ...,
                [ 93,  86,  91, ...,  66,  74,  78],
                [106,  99,  99, ...,  64,  72,  70],
                [ 69,  81, 101, ...,  60,  67,  58]], dtype=uint8)
```

```
In [46]: img1_red[:, :, 2]
```

```
Out[46]: array([[ 93,  93,  93, ..., 232, 232, 232],
                [ 93,  93,  94, ..., 232, 232, 232],
                [ 94,  94,  95, ..., 232, 232, 232],
                ...,
                [ 99,  92,  97, ...,  71,  79,  83],
                [112, 105, 105, ...,  69,  77,  75],
                [ 75,  87, 107, ...,  67,  74,  65]], dtype=uint8)
```

```
In [48]: img1_red[:, :, 1] = 0
```

```
In [49]: img1_red[:, :, 1]
```

```
Out[49]: array([[0, 0, 0, ..., 0, 0, 0],
                [0, 0, 0, ..., 0, 0, 0],
                [0, 0, 0, ..., 0, 0, 0],
                ...,
                [0, 0, 0, ..., 0, 0, 0],
                [0, 0, 0, ..., 0, 0, 0],
                [0, 0, 0, ..., 0, 0, 0]], dtype=uint8)
```

```
In [50]: plt.imshow(img1_red)
```

```
Out[50]: <matplotlib.image.AxesImage at 0x2d3cb583860>
```



```
In [51]: img1_red[:, :, 2]
```

```
Out[51]: array([[ 93,  93,  93, ..., 232, 232, 232],
                [ 93,  93,  94, ..., 232, 232, 232],
                [ 94,  94,  95, ..., 232, 232, 232],
                ...,
                [ 99,  92,  97, ...,  71,  79,  83],
                [112, 105, 105, ...,  69,  77,  75],
                [ 75,  87, 107, ...,  67,  74,  65]], dtype=uint8)
```

```
In [52]: img1_red[:, :, 2] = 0
```

```
In [53]: plt.imshow(img1_red)
```

```
Out[53]: <matplotlib.image.AxesImage at 0x2d3cb5e3560>
```



```
In [54]: img1_arr
```

```

Out[54]: array([[ 85,  94,  93],
                [ 85,  94,  93],
                [ 85,  94,  93],
                ...,
                [234, 234, 232],
                [234, 234, 232],
                [234, 234, 232]],

                [[ 85,  94,  93],
                 [ 85,  94,  93],
                 [ 86,  95,  94],
                 ...,
                 [234, 234, 232],
                 [234, 234, 232],
                 [234, 234, 232]],

                [[ 86,  95,  94],
                 [ 86,  95,  94],
                 [ 87,  96,  95],
                 ...,
                 [234, 234, 232],
                 [234, 234, 232],
                 [234, 234, 232]],

                ...,

                [[ 86,  93,  99],
                 [ 79,  86,  92],
                 [ 84,  91,  97],
                 ...,
                 [ 57,  66,  71],
                 [ 65,  74,  79],
                 [ 69,  78,  83]],

                [[ 99, 106, 112],
                 [ 92,  99, 105],
                 [ 92,  99, 105],
                 ...,
                 [ 55,  64,  69],
                 [ 63,  72,  77],
                 [ 61,  70,  75]],

                [[ 62,  69,  75],
                 [ 74,  81,  87],
                 [ 94, 101, 107],
                 ...,
                 [ 51,  60,  67],
                 [ 58,  67,  74],
                 [ 49,  58,  65]]], dtype=uint8)

```

```
In [55]: img1
```


Out[55]:

In [56]: `arr1 = np.asarray(img1)`In [57]: `arr1`


```

Out[57]: array([[ 85,  94,  93],
                [ 85,  94,  93],
                [ 85,  94,  93],
                ...,
                [234, 234, 232],
                [234, 234, 232],
                [234, 234, 232]],

                [[ 85,  94,  93],
                 [ 85,  94,  93],
                 [ 86,  95,  94],
                 ...,
                 [234, 234, 232],
                 [234, 234, 232],
                 [234, 234, 232]],

                [[ 86,  95,  94],
                 [ 86,  95,  94],
                 [ 87,  96,  95],
                 ...,
                 [234, 234, 232],
                 [234, 234, 232],
                 [234, 234, 232]],

                ...,

                [[ 86,  93,  99],
                 [ 79,  86,  92],
                 [ 84,  91,  97],
                 ...,
                 [ 57,  66,  71],
                 [ 65,  74,  79],
                 [ 69,  78,  83]],

                [[ 99, 106, 112],
                 [ 92,  99, 105],
                 [ 92,  99, 105],
                 ...,
                 [ 55,  64,  69],
                 [ 63,  72,  77],
                 [ 61,  70,  75]],

                [[ 62,  69,  75],
                 [ 74,  81,  87],
                 [ 94, 101, 107],
                 ...,
                 [ 51,  60,  67],
                 [ 58,  67,  74],
                 [ 49,  58,  65]]], dtype=uint8)

```

```
In [58]: type(arr1)
```

```
Out[58]: numpy.ndarray
```

```
In [60]: arr1.shape
```

```
Out[60]: (183, 276, 3)
```

```
In [61]: plt.imshow(arr1)
```

Out[61]: <matplotlib.image.AxesImage at 0x2d3cb93c9e0>



In [62]: `img2 = arr1.copy()`

In [63]: `img2[:, :, 0] = 0`

In [64]: `plt.imshow(img2)`

Out[64]: <matplotlib.image.AxesImage at 0x2d3cb1dc380>



In [65]: `img2[:, :, 1]`

```
Out[65]: array([[ 94,  94,  94, ..., 234, 234, 234],
                [ 94,  94,  95, ..., 234, 234, 234],
                [ 95,  95,  96, ..., 234, 234, 234],
                ...,
                [ 93,  86,  91, ...,  66,  74,  78],
                [106,  99,  99, ...,  64,  72,  70],
                [ 69,  81, 101, ...,  60,  67,  58]], dtype=uint8)
```

```
In [71]: img2[:, :, 1]
```

```
Out[71]: array([[0, 0, 0, ..., 0, 0, 0],
                [0, 0, 0, ..., 0, 0, 0],
                [0, 0, 0, ..., 0, 0, 0],
                ...,
                [0, 0, 0, ..., 0, 0, 0],
                [0, 0, 0, ..., 0, 0, 0],
                [0, 0, 0, ..., 0, 0, 0]], dtype=uint8)
```

```
In [72]: img2[:, :, 1]=0
```

```
In [74]: plt.imshow(img2)
```

```
Out[74]: <matplotlib.image.AxesImage at 0x2d3cba45820>
```



```
In [ ]:
```